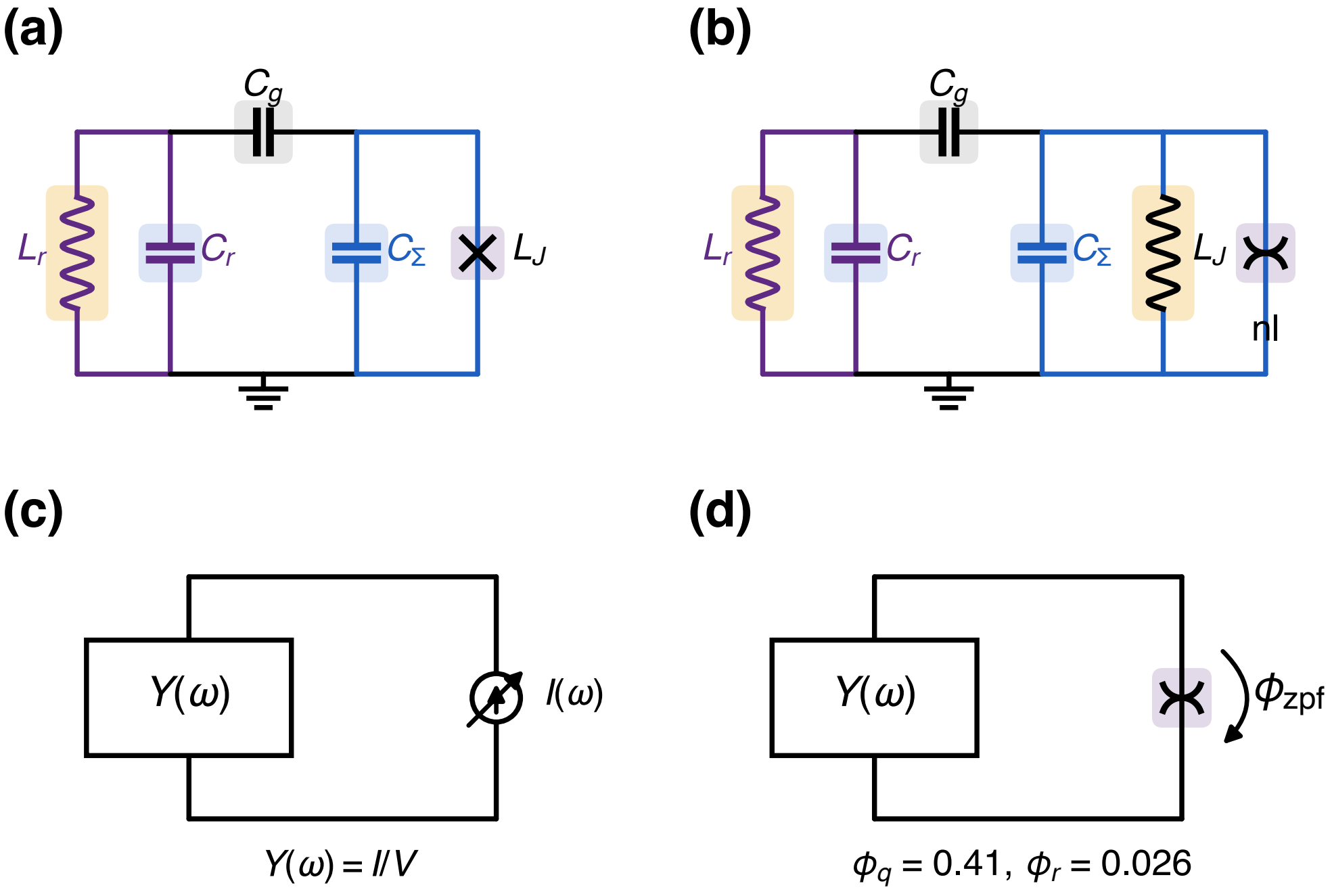
(a)
C_g
L_r
C_r
C_Σ
L_J
(b)
C_g
L_r
C_r
C_Σ
L_J
nl
(c)
$Y(\omega)$
$I(\omega)$
$Y(\omega) = I/V$
(d)
$Y(\omega)$
ϕ_{zpf}
$\phi_q = 0.41,\ \phi_r = 0.026$